THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $5x + 12$	$3x + 12 + 2x$
2	**(a)** $x^2 + 7x + 10$	Multiply term by term
3	**(a)** 15	$3 \times 9 - 5 \times 4 + 8$
4	**(b)** $10x + 14$	Perimeter $= 2(l + b)$
5	**(b)** 53	$(x + y)^2 = x^2 + y^2 + 2xy$
6	**(b)** $6n + 6$	$2n + (2n + 2) + \ldots$: $6n + 6$
7	**(a)** $9x + 10$	$5x + 10 + 4x$
8	**(b)** $x^2 + 7x + 12$	Multiply term by term
9	**(d)** 40	$4 \times 12 - 6 \times 3 + 10$
10	**(c)** $14x + 14$	Perimeter $= 2(l + b)$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2x + 10 + 3x - 3$ (1); $5x + 7$ (1).	1+1
12	$x^2 + 9x + 4x + 36$ (1); $x^2 + 13x + 36$ (1).	1+1
13	$(10a + b) - (10b + a) = 9(a - b)$ (1) $= 9 \times 2 = 18$ (1).	1+1
14	$4x + 12 + 2x - 12$ (1); $6x$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter $9x + 9$ cm (2); for $x = 2$: $9 \times 2 + 9 = 27$ cm (1).	2+1
16	$x^2 + y^2 = (x + y)^2 - 2xy$ (1) $= 11^2 - 2 \times 28$ (1) $= 65$ (1).	1+1+1
17	Terms $2n, 2n + 2, \ldots, 2n + 4$ (1); sum $6n + 6$ (1); for $n = 4$ the numbers add to 30 and 30 agrees (1).	1+1+1
	OR: Perimeter $10x + 12$ cm (2); for $x = 3$: $10 \times 3 + 12 = 42$ cm (1).	2+1
18	$x^2 + y^2 = (x + y)^2 - 2xy$ (1) $= 13^2 - 2 \times 40$ (1) $= 89$ (1).	1+1+1
	OR: Terms $2n, 2n + 2, \ldots, 2n + 8$ (1); sum $10n + 20$ (1); for $n = 3$ the numbers add to 50 and 50 agrees (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2 + 8x + 15 - x^2 - 4x$ (2); $4x + 15$ (1); at $x = 2$: 23 (2).	2+1+2
	OR: $(10a + b) + (10b + a) = 11a + 11b = 11(a + b)$ (3); for 3 and 9: 132 (1); divisible by 11 (1).	3+1+1
20	$x^2 + 9x + 14 - x^2 - 3x$ (2); $6x + 14$ (1); at $x = 4$: 38 (2).	2+1+2
	OR: $(10a + b) + (10b + a) = 11a + 11b = 11(a + b)$ (3); for 4 and 8: 132 (1); divisible by 11 (1).	3+1+1

Mistakes to avoid:

- Adding unlike terms: $3x + 4x^2 = 7x^3$; only like terms are combined, and $3x + 4x = 7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a + b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x - 3) = -x + 3$, not $-x - 3$.
- Writing $(a + b)^2 = a^2 + b^2$ or $(a - b)^2 = a^2 - b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n + 1$ for $n = 1, 2, 3$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 58	$(x+y)^2 = x^2 + y^2 + 2xy$
2	**(d)** $8n + 12$	$2n + (2n+2) + \ldots$: $8n + 12$
3	**(d)** $7x + 24$	$4x + 24 + 3x$
4	**(c)** $x^2 + 8x + 12$	Multiply term by term
5	**(d)** 45	$6 \times 8 - 2 \times 5 + 7$
6	**(c)** $12x + 14$	Perimeter $= 2(l + b)$
7	**(c)** 74	$(x+y)^2 = x^2 + y^2 + 2xy$
8	**(b)** $10n + 20$	$2n + (2n+2) + \ldots$: $10n + 20$
9	**(b)** $11x + 18$	$6x + 18 + 5x$
10	**(d)** $x^2 + 10x + 21$	Multiply term by term

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2 + 8x + 3x + 24$ (1); $x^2 + 11x + 24$ (1).	1+1
12	$(10a + b) - (10b + a) = 9(a - b)$ (1) $= 9 \times 3 = 27$ (1).	1+1
13	$5x + 10 + 4x - 12$ (1); $9x - 2$ (1).	1+1
14	$x^2 + 6x + 5x + 30$ (1); $x^2 + 11x + 30$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter $9x + 6$ cm (2); for $x = 5$: $9 \times 5 + 6 = 51$ cm (1).	2+1
16	$x^2 + y^2 = (x+y)^2 - 2xy$ (1) $= 8^2 - 2 \times 15$ (1) $= 34$ (1).	1+1+1
17	Terms $2n, 2n+2, \ldots, 2n+6$ (1); sum $8n + 12$ (1); for $n = 6$ the numbers add to 60 and 60 agrees (1).	1+1+1
	OR: Perimeter $11x + 11$ cm (2); for $x = 4$: $11 \times 4 + 11 = 55$ cm (1).	2+1
18	$x^2 + y^2 = (x+y)^2 - 2xy$ (1) $= 14^2 - 2 \times 33$ (1) $= 130$ (1).	1+1+1
	OR: Terms $2n, 2n+2, \ldots, 2n+10$ (1); sum $12n + 30$ (1); for $n = 2$ the numbers add to 54 and 54 agrees (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2 + 10x + 24 - x^2 - 5x$ (2); $5x + 24$ (1); at $x = 3$: 39 (2).	2+1+2
	OR: $(10a + b) + (10b + a) = 11a + 11b = 11(a + b)$ (3); for 5 and 6: 121 (1); divisible by 11 (1).	3+1+1
20	$x^2 + 10x + 16 - x^2 - 6x$ (2); $4x + 16$ (1); at $x = 5$: 36 (2).	2+1+2
	OR: $(10a + b) + (10b + a) = 11a + 11b = 11(a + b)$ (3); for 2 and 8: 110 (1); divisible by 11 (1).	3+1+1

Mistakes to avoid:

- Adding unlike terms: $3x + 4x^2 = 7x^3$; only like terms are combined, and $3x + 4x = 7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a + b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x - 3) = -x + 3$, not $-x - 3$.
- Writing $(a+b)^2 = a^2 + b^2$ or $(a-b)^2 = a^2 - b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n + 1$ for $n = 1, 2, 3$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 17	$2 \times 15 - 9 \times 2 + 5$
2	**(a)** $16x + 18$	Perimeter $= 2(l + b)$
3	**(b)** 25	$(x + y)^2 = x^2 + y^2 + 2xy$
4	**(c)** $12n + 30$	$2n + (2n + 2) + \ldots$: $12n + 30$
5	**(d)** $16x + 14$	$7x + 14 + 9x$
6	**(a)** $x^2 + 13x + 40$	Multiply term by term
7	**(b)** 34	$5 \times 10 - 4 \times 7 + 12$
8	**(b)** $16x + 16$	Perimeter $= 2(l + b)$
9	**(b)** 137	$(x + y)^2 = x^2 + y^2 + 2xy$
10	**(d)** $14n + 42$	$2n + (2n + 2) + \ldots$: $14n + 42$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(10a + b) - (10b + a) = 9(a - b)$ (1) $= 9 \times 4 = 36$ (1).	1+1
12	$3x + 21 + 6x - 12$ (1); $9x + 9$ (1).	1+1
13	$x^2 + 11x + 2x + 22$ (1); $x^2 + 13x + 22$ (1).	1+1
14	$(10a + b) - (10b + a) = 9(a - b)$ (1) $= 9 \times 5 = 45$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Perimeter $10x + 12$ cm (2); for $x = 6$: $10 \times 6 + 12 = 72$ cm (1).	2+1
16	$x^2 + y^2 = (x + y)^2 - 2xy$ (1) $= 16^2 - 2 \times 55$ (1) $= 146$ (1).	1+1+1
17	Terms $2n, 2n + 2, \ldots, 2n + 12$ (1); sum $14n + 42$ (1); for $n = 5$ the numbers add to 112 and 112 agrees (1).	1+1+1
	OR: Perimeter $11x + 10$ cm (2); for $x = 7$: $11 \times 7 + 10 = 87$ cm (1).	2+1
18	$x^2 + y^2 = (x + y)^2 - 2xy$ (1) $= 17^2 - 2 \times 60$ (1) $= 169$ (1).	1+1+1
	OR: Terms $2n, 2n + 2, \ldots, 2n + 4$ (1); sum $6n + 6$ (1); for $n = 9$ the numbers add to 60 and 60 agrees (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2 + 10x + 25 - x^2 - 2x$ (2); $8x + 25$ (1); at $x = 6$: 73 (2).	2+1+2
	OR: $(10a + b) + (10b + a) = 11a + 11b = 11(a + b)$ (3); for 7 and 9: 176 (1); divisible by 11 (1).	3+1+1
20	$x^2 + 12x + 27 - x^2 - 7x$ (2); $5x + 27$ (1); at $x = 2$: 37 (2).	2+1+2
	OR: $(10a + b) + (10b + a) = 11a + 11b = 11(a + b)$ (3); for 1 and 9: 110 (1); divisible by 11 (1).	3+1+1

Mistakes to avoid:

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- Writing the two-digit number as ab (product) instead of $10a + b$.
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- Sign mistakes when removing brackets: $-(x - 3) = -x + 3$, not $-x - 3$.
- Writing $(a + b)^2 = a^2 + b^2$ or $(a - b)^2 = a^2 - b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n + 1$ for $n = 1, 2, 3$).

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